Demo: Osphra: Finding Odor Sources without Mapping Every Point

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Abstract

Finding an odor source indoors usually means sampling every spot on a grid and picking out the position with the strongest reading. We present Osphra, a handheld electronic nose that makes the search selective. An ESP32-S3 reads the BME688 gas sensor, and Samsung Galaxy A16 smartphone AR tracking gives every reading a position. Partial measurements decide where to sample next; a repeated stabilized response decides when to stop. A full grid sweep took 36 probes and 267 s, while guided search reached the source in 3 probes (30 s) and 6 probes (78 s).

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**; • **Computer systems organization** → *Embedded systems.*

Keywords

electronic nose, mobile sensing, VOC, source localization, augmented reality, Bluetooth Low Energy

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1 Introduction

Everyday life is full of moments when knowing *where* a smell comes from matters: a faint gas odor in the kitchen, food spoiling somewhere in a packed refrigerator. However, the only search tool most people have is their own nose: people are often poor at telling where an odor comes from[3], and the longer the search goes on, the less they can smell, as the nose adapts to a persistent odor[4]. A smartphone that can track and localize odors would turn each of these from a frustrating hunt into a quick measurement task.

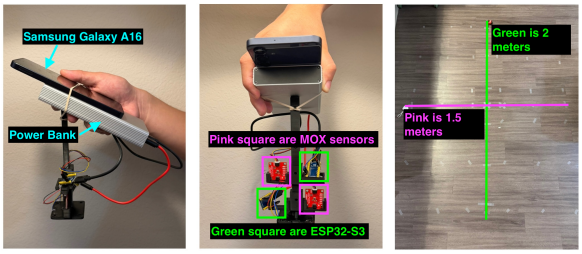


Figure 1: The Osphra handheld prototype (left); its underside with MOX sensors and ESP32-S3 boards (middle); the marked indoor search area (right).

Low-cost volatile organic compound (VOC) sensors are now portable, and smartphones track their own position in real time, so in principle every gas reading can carry a location, and readings should sharpen near the source. Turning this into a working search tool raises three challenges. First, a metal-oxide (MOX) gas sensor is slow: its response delays by seconds and recovers even more slowly, so a reading taken while walking describes a spot the user has already left, and the system must decide which readings actually belong to where they were taken. Second, tracing a source works by comparison. At each step the search asks one question: is the air here stronger than the air just visited? But drift breaks this comparison. Sensor resistance shifts with temperature, humidity, and recent exposure, so a rise from one spot to the next may be nothing but drift, and a search that operates on raw readings would follow the sensor rather than the actual gas. Third, the search must end without a map: directional guidance may still be consistent even when no source is present, so the system must decide, from chemistry rather than geometry, when the source is actually found.

We present Osphra, a handheld prototype that turns off-the-shelf VOC sensing and phone positioning into a human-in-the-loop search tool by answering these three challenges

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in turn. First, a reading counts only after the user stops and the signal stabilizes, so each spatial point belongs to the spot where it was taken. Second, the app subtracts a running baseline before comparing readings, so the on-screen arrow that suggests the next measurement follows the gas, not the drift. Third, the search ends only on repeated chemistry: a strong response must recur at the same spot before the app declares the source found. At the booth, one attendee hides a household VOC source at one of three marked spots, and another follows the arrow until Osphra reports FOUND.

2 System and Method

Osphra has two parts (Fig. 1): a battery-powered sensing node and a phone app. The node is a Bosch BME688 metal-oxide (MOX) gas sensor [1] on an ESP32-S3, streaming gas resistance over Bluetooth at ~4 samples/s. The method is one loop (Fig. 2), run once per measurement; its three stages answer the three challenges of the introduction in order, and Section 3 shows the loop running live.

Accepting a measurement. The first problem is that the sensor is slower than the user: on our rig the response lags by 1.1–2.7 s and takes over 4 s to recover, so a reading taken while walking belongs to a position the user has already left. Osphra therefore accepts readings only when the phone is still. ARCore [2] tracks the phone in a floor frame anchored by a printed marker and stamps every reading with its position. Readings taken while moving are set aside (TRANSIT); after the user stops, the app waits 2 s (SETTLING), then takes the median over 3 s (STABLE) as one spatial point.

Choosing the next spot. The second problem is that each point costs a 5 s stop, so the app must choose directions well. It works with relative strength rather than raw readings: with $v_t = \log_{10} R_t$ for resistance R_t, strength is $s_t = b_t - v_t$, where b_t is the median of v over the preceding 90 s; subtracting this running baseline cancels the slow drift of the sensor, so s_t says "stronger than the air just visited," not how much gas is present. The app fits a local plane $s_i \approx \alpha + \mathbf{g}^{\mathsf{T}}(\mathbf{p}_i - \mathbf{p}_0)$ to the points near the current position $\mathbf{p}_0$, and the arrow points uphill along $\mathbf{g}$, which is shown only when the nearby points spread in two dimensions, the slope is at least twice its standard error, and the sensor is not recovering. Accepted directions fuse into a source hint. The arrow changes where the user measures next; it never changes what counts as a valid observation.

Confirming the source. The last problem is that direction alone cannot confirm a source: fused directions may agree even when no strong source is present, so the search ends only on repeated chemistry. A stable point becomes CANDIDATE when its response stands clearly above the local noise; FOUND requires a second strong point within 20 cm at least

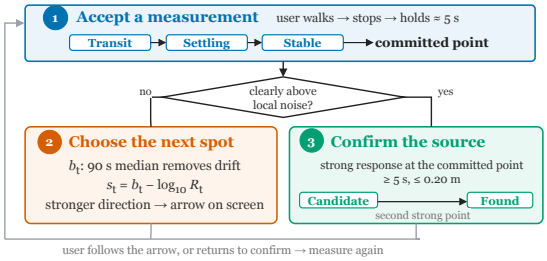


Figure 2: The Osphra loop: each committed reading either updates the guidance arrow or, when a strong response recurs at the same spot, confirms the source.

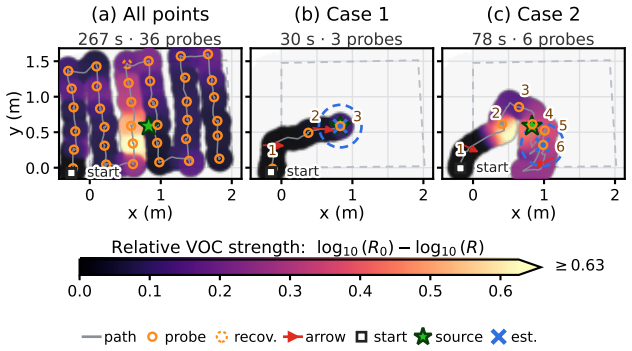


Figure 3: Exhaustive grid mapping (a) versus two guided searches (b, c) of the same 1.5×2.0 m zone on one shared, measured-only color scale. Panel headers give each run's total time and committed probes; both guided runs commit a probe within 20 cm of the recorded source (★), while the raster has no stopping rule before its final probe.

5 s later, after the user leaves, lets the sensor recover, and returns. Geometry guides the search; repeatable chemistry locates the source.

3 Demonstration

Attendees run the loop from Section 2 themselves: they take a reading at a spot, watch the system turn it into a direction, and follow the arrow to the next spot, repeating until a second strong response at the same place confirms the source.

Setup. One attendee hides an organizer-approved household VOC source at one of three identical marked spots. Another attendee, who does not see the choice, starts from a marked origin holding the phone and the sensing node.

The search loop. The searcher walks to a spot and stands still. After the 5 s point protocol completes, a colored dot appears on the live map at that position. A warmer color and a larger strength value mean a stronger response. Once enough points surround the searcher, an arrow appears and

points to the next direction. The searcher follows it, stops, and repeats. Each accepted point visibly redraws the arrow and the source estimate, so the audience watches the search narrow instead of filling a grid.

Confirming the source. When a point stands clearly above the local noise, the state changes to CANDIDATE and the app instructs the searcher to step away while the sensor recovers. The searcher then returns and takes one more point at the same spot, and a second strong response turns the state to FOUND. Figure 3 shows two guided runs beside the full sweep: one heads almost straight to the source, one recovers from a detour, and both end with a committed probe within 20 cm of it.

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